

☒ 1. Protecting Information System Security

☒ Definition:

Protecting Information System (IS) security involves implementing safeguards to ensure the **confidentiality, integrity, and availability (CIA)** of data and systems.

☒ Key Concepts:

- **Confidentiality:** Ensuring only authorized users can access data.
- **Integrity:** Ensuring data is accurate and unaltered.
- **Availability:** Ensuring systems and data are accessible when needed.

☒ Protection Techniques:

- **Authentication & Authorization:** Ensuring only legitimate users can access the system.
- **Access Control Mechanisms:** DAC, MAC, RBAC.
- **Firewalls and Intrusion Detection/Prevention Systems (IDS/IPS).**
- **Encryption:** Data in transit and at rest.
- **Backup and Disaster Recovery Plans.**
- **Patch Management:** Fixing vulnerabilities through updates.

☒ Physical Security:

- Surveillance, biometric access, and environmental controls (fire alarms, temperature).
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☒ 2. Security in Mobile and Wireless Computing

☒ Mobile Security:

- Mobile devices are **vulnerable** due to portability, always-on connectivity, and app-based ecosystem.

☒ Mobile Threats:

- **Malware** via third-party apps.
- **Unsecured Wi-Fi** networks.
- **Data leakage** through app permissions.
- **Lost/stolen** devices.

☒ Protection Mechanisms:

- **Mobile Device Management (MDM)**.

- Enforcing **encryption** and **remote wipe** policies.
- Secure app development practices.
- Two-Factor Authentication (2FA) for mobile access.

☒ Wireless Security:

- Wireless networks (Wi-Fi, Bluetooth) are susceptible to attacks like:
 - Eavesdropping (sniffing)
 - Man-in-the-middle (MITM)
 - Rogue Access Points

☒ Protection Techniques:

- WPA3 over WEP/WPA2.
 - VPNs for secure communications.
 - MAC address filtering and SSID hiding.
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☒ 3. Credit Card Frauds in Mobile and Wireless Computing

☒ Common Types of Credit Card Frauds:

1.
Phishing: Deceiving users to provide card details via fake apps/sites.
2.
Skimming: Cloning card data via compromised readers.
3.
Data breaches: Attacks on mobile wallets/apps to steal payment data.
4.
SIM swap: Hijacking mobile number to receive OTPs.

☒ Mobile-Specific Threats:

- Malware that intercepts SMS-based OTPs.
- Fake mobile wallets or apps.
- Public Wi-Fi data sniffing.

☒ Prevention:

- Use **tokenization** for transactions.
- **EMV chip cards** over magnetic stripe.

- App vetting and sandboxing on mobile.
 - Enforce PCI-DSS standards for app-based payment processing.
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☒ 4. Information Security Management (ISM)

☒ Definition:

ISM is the process of managing information security using frameworks, policies, procedures, and controls.

☒ Frameworks & Standards:

- ISO/IEC 27001: International standard for ISMS.
- NIST: Cybersecurity Framework used in US organizations.

☒ Key Processes:

1. Risk Assessment: Identify and evaluate security risks.
2. Risk Treatment: Implement controls to mitigate risks.
3. Security Policy Development.
4. Training & Awareness.

5.
Incident Management and Response.
 6.
Continuous Monitoring and Improvement.
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☒ 5. Fundamentals of Information Security

☒ Core Principles (CIA Triad):

1.
Confidentiality – Protecting data from unauthorized access.
2.
Integrity – Ensuring accuracy and reliability of data.
3.
Availability – Ensuring timely access to resources.

☒ Security Mechanisms:

- **Cryptography**: Ensures confidentiality and integrity.
- **Access Controls**: Ensures only authorized access.
- **Security Policies**: Sets rules for users and systems.
- **Incident Response**: Deals with breaches and recovery.

☒ Importance:

- Prevents data breaches, legal liabilities, and financial loss.
- Builds trust with customers and stakeholders.
- Ensures compliance with regulations like GDPR, HIPAA, PCI-DSS.